Project: **Explore the addition of a new concept to a diffusion model**
Base model: **Stable Diffusion v1.5**
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1. Introduction

Diffusion models can generate high-quality images from text prompts, but they often struggle with concepts that are rare, private, or simply absent from the training data. In this project, I explored how to introduce a new visual concept into a publicly available pretrained diffusion model using three standard personalization techniques: Textual Inversion, LoRA, and DreamBooth.

The main objective was to teach the model a novel concept using a small dataset and compare the approaches in terms of training stability, speed, and image quality. Additionally, I conducted a small set of experiments to understand which factors most affect the final results (e.g., the number of training images, LoRA rank, and training the text encoder).

2. Concept description

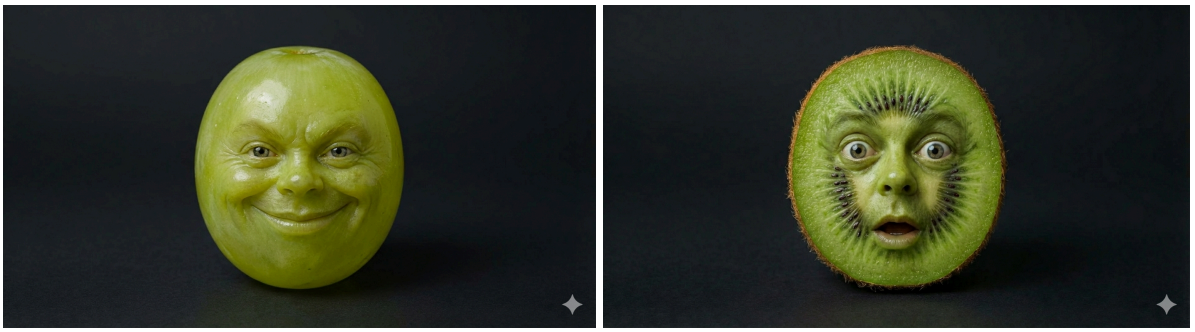
Concept name: fruitface

Target concept: a realistic fruit/berry where a single human face is physically part of the fruit (embedded/carved into the skin), preserving the fruit's texture, color, and geometry.

Important failure modes to avoid:

- **a person holding a fruit**
- **a face behind a fruit / a human portrait in the background**

Examples of correct pictures (training set):



3. Data preparation (Custom)

I used a small custom dataset collected manually (**10 images generated by Gemini NanoBanana Pro**). The images were preprocessed into 512×512 resolution and stored in a dedicated folder for training. The dataset focuses on the fruitface concept under slightly different viewpoints and lighting to improve generalization.

Dataset summary:

- **Number of instance images: 10**
- **Resolution: 512×512**
- **Base concept token: `sksfruitface` (LoRA/DreamBooth), `<fruitface>` (Textual Inversion)**

Why this dataset is challenging:

- **The model tends to interpret "human face + fruit" as "person behind fruit" or "person holding fruit." Therefore, dataset consistency and prompt design are critical to enforce the face being part of the fruit.**
- **Example of wrong generation:**



4. Methodology (hyperparameters + training setup)

4.1 Training setup (Baseline)

- **Resolution: 512×512**
- **Train batch size: 1**
- **Gradient accumulation: 1**
- **Scheduler: constant**
- **Warmup: 0**
- **Seed: 123 (for training and for generation when comparing results)**
- **Logging: TensorBoard (`report_to=tensorboard`)**

4.2 Textual Inversion (TI)

Goal: learn a new token embedding for `<fruitface>`.

Training:

- `--placeholder_token "<fruitface>"`
- `--initializer_token "fruit"`
- `--learnable_property "object"`
- `--resolution 512`
- `--train_batch_size 1`
- `--max_train_steps 1200`

- `--learning_rate 5e-4`
- `--checkpointing_steps 200`

Validation during training:

- `--validation_prompt "a photo of <fruitface> fruit with a cute human face, studio lighting, high detail"`
- `--validation_steps 200`
- `--num_validation_images 4`

Notes / issues:
Textual Inversion is lightweight and fast, but capacity is limited. In my case it sometimes generated “fruit + human in background” instead of an embedded face (prompt sensitivity).

4.3 LoRA

Goal: train LoRA adapters for the concept token `sksfruitface`.

Training:

- `--rank 8`
- `--mixed_precision bf16`
- `--resolution 512`

- `--max_train_steps 1200`
- `--learning_rate 1e-4`
- `--checkpointing_steps 200`

Validation during training:

- `--validation_prompt "a photo of sksfruitface fruit with a cute human face, studio lighting, high detail"`
- `--validation_epochs 1`
- `--num_validation_images 4`

Notes / issues:
LoRA trained quickly and produced the most controllable results among the lightweight methods. The main failure mode was the model interpreting “human face” as a separate person; this was mitigated by prompt/negative prompt design.

4.4 DreamBooth

Goal: finetune full model weights while preserving the base class “fruit”.

Training:

- `--with_prior_preservation`
- `--prior_loss_weight 1.0`

- `--num_class_images 60` (class prompt: `"a photo of fruit"`)
- `--resolution 512`
- `--max_train_steps 400`
- `--learning_rate 2e-6`
- `--mixed_precision bf16`
- `--use_8bit_adam`
- `--checkpointing_steps 400`
- Instance prompt: `"a photo of sksfruitface fruit with a cute human face"`
- Class prompt: `"a photo of fruit"`

Notes / issues:
DreamBooth has the highest capacity, but is heavier to train and more prone to overfitting with small datasets. It can lock onto the concept strongly, but sometimes at the cost of diversity.

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5. Example generations

To compare methods fairly, I generated images using the same set of prompts and fixed seeds.

Evaluation prompts (examples):

1. `a photo of sksfruitface fruit with a cute human face, studio lighting, high detail`

2. `sksfruitface strawberry with a cute human face, macro photo, shallow depth of field`

Negative prompt (used for quality control):
`blurry, low quality, deformed, bad anatomy`

- **Figure 1: Baseline (no training)**



Figure 2: Textual Inversion results (`TI_base`)



Figure 3: LoRA results (`LORA_base_r8`)



Figure 4: DreamBooth full results (`DB_full_fast`)



Qualitative summary:

- **Textual Inversion frequently produced a *human subject near a fruit* or a *face as a separate element*, so the "embedded face" constraint was often violated.**

- **LoRA learned the concept more reliably and produced multiple correct "face-on-fruit-surface" generations.**

- **DreamBooth full most consistently produced the intended geometry: the face looks physically integrated into the fruit while preserving fruit texture and shape.**

6. Overall comparison

Textual Inversion: fastest and smallest artifact, but limited capacity for enforcing strong compositional constraints; often ambiguous for this concept.

LoRA: best speed–quality tradeoff; stable training and strong concept injection; easy to swap and reuse.

DreamBooth full: strongest concept binding and most consistent geometry, but heavier training and higher overfitting risk with small datasets; may reduce diversity.

7. Experiments (3 research tasks)

Experiment 1 — Effect of number of training images (n=3 vs n=5 vs n=10)

Question: how does dataset size affect concept quality and overfitting?

Setup: trained LoRA (rank=8) on three subsets:

- `n=3`
- `n=5`
- `n=10`

Common settings:

- **steps: 450**
- **lr: 1e-4**
- **resolution: 512**
- **seed: 123**

Results:

- `n=3`**: concept appears, but unstable (composition errors, identity drift, more “person behind fruit” artifacts).**



- **n=5: noticeably more stable; still occasional failures.**



- `n=10`: best balance: concept more consistent and looks more “embedded” into the fruit surface.



Conclusion: increasing the number of instance images improved stability and reduced spurious “person in background / separate face” failures.

Experiment 2 — LoRA rank (r=4 vs r=16)

Question: how does LoRA rank affect quality vs overfitting?

Setup: trained two LoRAs with same data, steps (450), lr (1e-4):

- `rank=4`
- `rank=16`

Results:

- **`r=4`: weaker concept strength, sometimes needs stronger prompts to activate (some composition errors).**



- **`r=16`: stronger concept injection (face appears more reliably, color matching), but higher chance of artifacts/overfitting if prompts are narrow.**



Conclusion: higher rank increased concept strength and activation reliability, but can increase artifacts/overfitting risk. In my runs, r=16 gave the strongest binding, while r=4 was weaker and more prompt-sensitive.

Experiment 3 — Training text encoder (ON vs OFF)

Question: does training text encoder help the token bind better?

Setup: LoRA rank=8, same data, steps=450:

- **Training text encoder OFF: (train only UNet LoRA)**
- **Training text encoder ON: (train UNet LoRA + text encoder)**

Results:

- **OFF: decent concept learning, but more prompt-sensitive; outputs frequently have softer (blurrier) fruit textures and less sharp edges, especially under more complex prompts.**



- **ON: token activation stronger and more consistent; however, some prompts showed slightly reduced variety.**



Conclusion: enabling text encoder training made token activation stronger and more consistent, but slightly reduced output diversity across the evaluation prompt set (some prompts became more similar).

8. Final summary.

Across my runs, LoRA provided the best speed–quality tradeoff and was the easiest method to iterate on. DreamBooth full produced the strongest and most consistent "face embedded into fruit" geometry, but with a higher risk of reduced diversity. Textual Inversion was the lightest method but struggled to enforce the compositional constraint.

Among the tested LoRA settings, increasing the number of instance images improved stability, higher rank increased concept strength and training the text encoder improved token activation consistency but sometimes reduced variation across prompts.